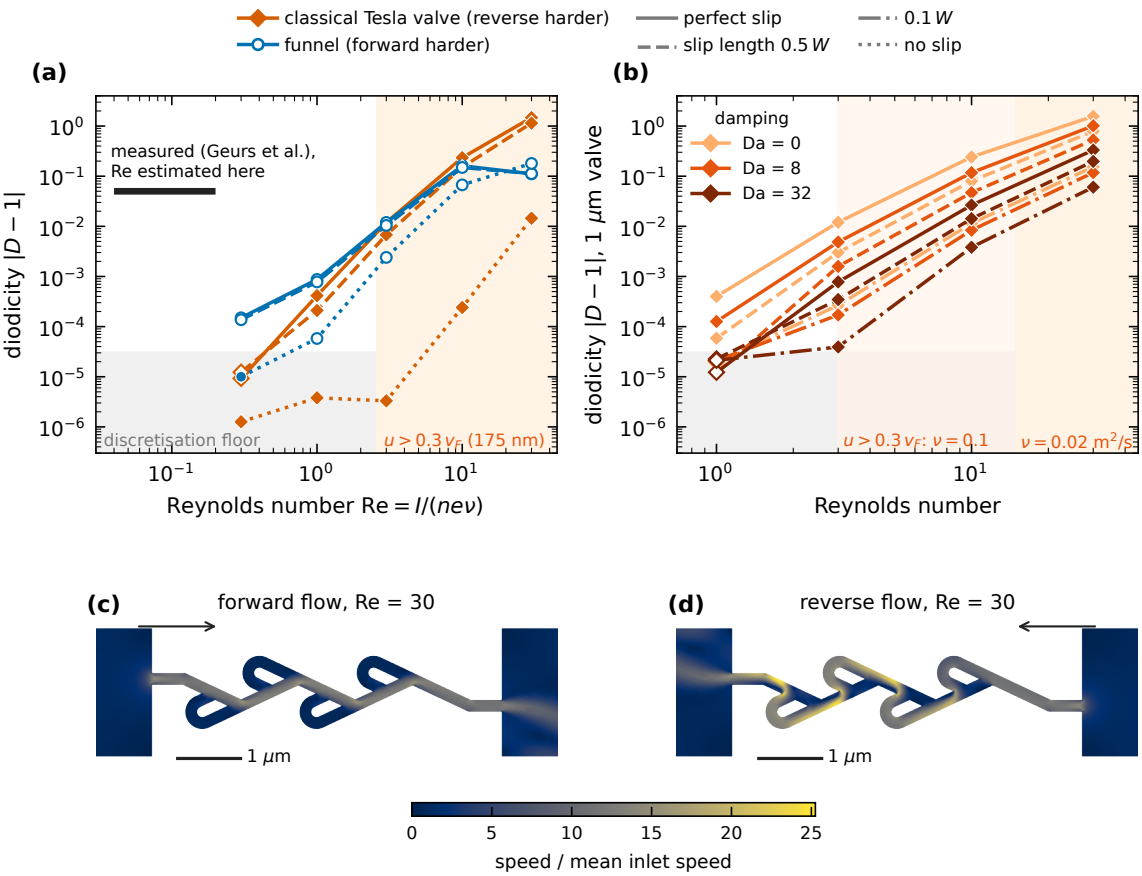

classical Tesla valve (reverse harder)
funnel (forward harder)
perfect slip
slip length 0.5 W
0.1 W
no slip
(a)
measured (Geurs et al.),
Re estimated here
diodicity |D − 1|
discretisation floor
u > 0.3 v_Fl (175 nm)
Reynolds number Re = l/(neν)
(b)
damping
Da = 0
Da = 8
Da = 32
diodicity |D − 1|, 1 μm valve
u > 0.3 v_F: ν = 0.1
ν = 0.02 m²/s
Reynolds number
(c)
forward flow, Re = 30
1 μm
(d)
reverse flow, Re = 30
1 μm
0
5
10
15
20
25
speed / mean inlet speed